MILITARY SPECIFICATION SHEET

PIN-RIVET, THREADED, TYPE II (100° FLUSH REDUCED SHEAR HEAD), CLASS 2
(6AL-4V TITANIUM, 95 KSI Fsu AND 160 KSI Ftu)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the pin-rivet, threaded described herein shall consist of this document and the issue in effect of MIL-P-87988.

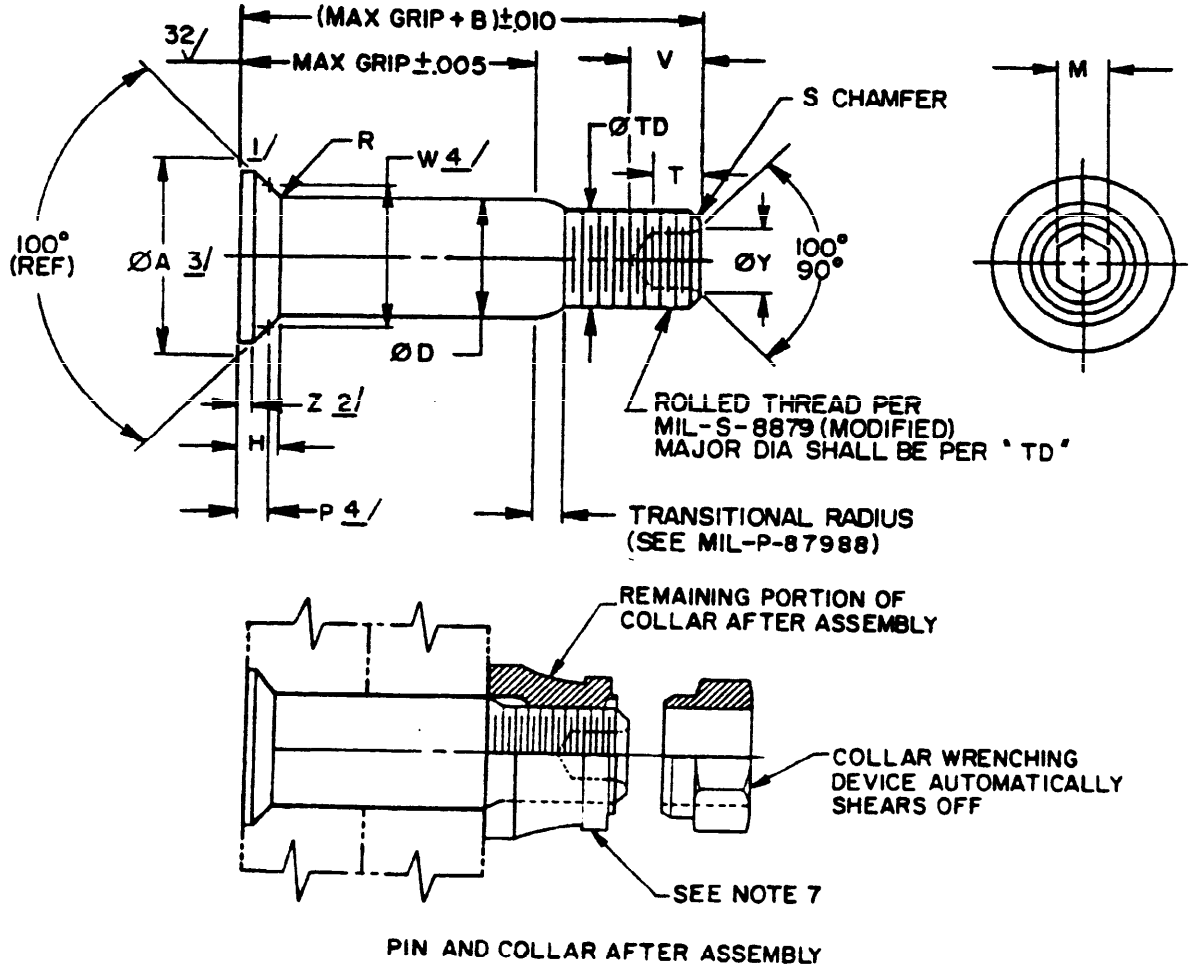


FIGURE 1. PIN-RIVET, THREADED, TYPE II (100° FLUSH REDUCED SHEAR HEAD), CLASS 2 (6AL-4V TITANIUM, 95 KSI Fsu AND 160 KSI Ftu)

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NOTES:

1. Dimensions in inches to be met after application of final coating or surface finish, unless otherwise specified.
1/ 2. Head edge out of roundness shall not exceed "F".
3. Concentricity: conical surface of head to "D" diameter within .005 FIM.
4. "H" dimensioned from maximum "D" diameter.
2/ 5. Curved or flat edge manufacturer's option.
5/ 6. Evidence of broken edge across points.
7. While this part is designed to be used in interference fit applications (because it has a transitional radius) it can also be used in noninterference or clearance applications. It is similar to HL11. "HL" is a registered trademark of Hi-Shear Corporation.
3/ 8. Concavity of head surface shall not exceed .001. Convexity of head surface shall not exceed .003 for -5, -6 and -8; .004 for -10 and -12; .006 for -14 and above (not including head marking identification).
9. Hole preparation in accordance with NAS618. (Information only)
4/ 10. All head dimensions shall comply with requirements of TABLE I of this slash sheet and TABLE X of procurement specification.

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TABLE I. DASH NUMBERS AND DIMENSIONS

DIA NO.	NOM SIZE	THREAD (MODIFIED)	ØA	B REF	ØD		ØTD	1/ F	H	R RAD	2/ Z MAX	S CHAMFER REF	SOCKET				DOUBLE SHEAR POUNDS MIN	TENSION POUNDS MIN
					WITHOUT COATING	WITH COATING							M HEX	T DEPTH	V MAX DRILL DEPTH	ØY		
5	5/32	.1640-32 UNJC-3A	.2612 .2564	.312	.1635 .1630	.1635 .1625	.1595 .1570	.004	.0410 .0390	.025 .015	.010	.031 x 37°	.0801 .0791	.100 .080	.140	5/	4,010	1,290
6	3/16	.1900-32 UNJF-3A	.3016 .2966	.325	.1895 .1890	.1895 .1885	.1840 .1810	.005	.0470 .0449	.030 .020	.015	.031 x 37°	.0806 .0791	.100 .080	.145	.119 .104	5,380	2,000
8	1/4	.2500-28 UNJF-3A	.3948 .3898	.395	.2495 .2490	.2495 .2485	.2440 .2410	.006	.0610 .0589	.030 .020	.015	.031 x 37°	.0967 .0947	.110 .090	.160	.142 .122	9,300	3,700
10	5/16	.3125-24 UNJF-3A	.4739 .4689	.500	.3120 .3115	.3120 .3110	.3060 .3020	.007	.0679 .0658	.040 .030	.015	.047 x 37°	.1295 .1270	.130 .110	.200	.180 .160	14,600	5,000
12	3/8	.3750-24 UNJF-3A	.5604 .5554	.545	.3745 .3740	.3745 .3735	.3680 .3640	.008	.0780 .0759	.040 .030	.015	.047 x 37°	.1617 .1582	.160 .140	.239	.217 .197	21,000	7,200
14	7/16	.4375-20 UNJF-3A	.6680 .6620	.635	.4370 .4365	.4370 .4360	.4310 .4260	.009	.0969 .0944	.050 .040	.022	.047 x 37°	.1930 .1895	.190 .170	.278	.253 .233	28,600	10,000
16	1/2	.5000-20 UNJF-3A	.7540 .7480	.685	.4995 .4990	.4995 .4985	.4930 .4880	.010	.1068 .1043	.050 .040	.022	.047 x 37°	.2242 .2207	.220 .200	.323	.289 .269	37,300	13,500

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REQUIREMENTS:

1. Material. 6A1 - 4V titanium alloy (UNS R56400) in accordance with AMS 4928 or AMS 4967.

2. Protective coating and surface treatment.

   A - Aluminum coated in accordance with MIL-C-85614 and lubricated with cetyl alcohol in accordance with MIL-L-87132 Type I, Grade B.

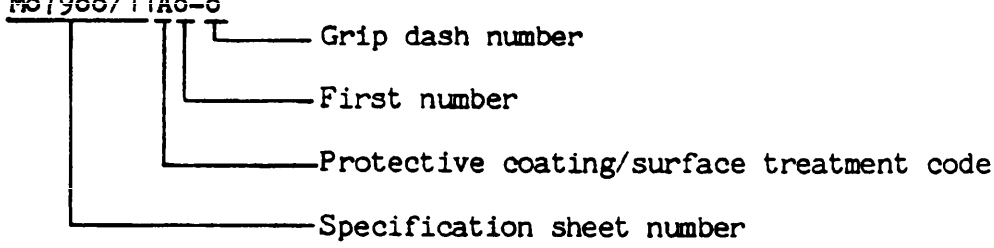
   C - Cetyl alcohol lubricant in accordance with MIL-L-87132 Type I, Grade B.

3. Heat treatment. 160,000 psi tensile minimum (95,000 psi shear minimum) in accordance with MIL-H-81200.

4. Surface texture. Unless otherwise specified, all surfaces shall have a 125 microinch finish.

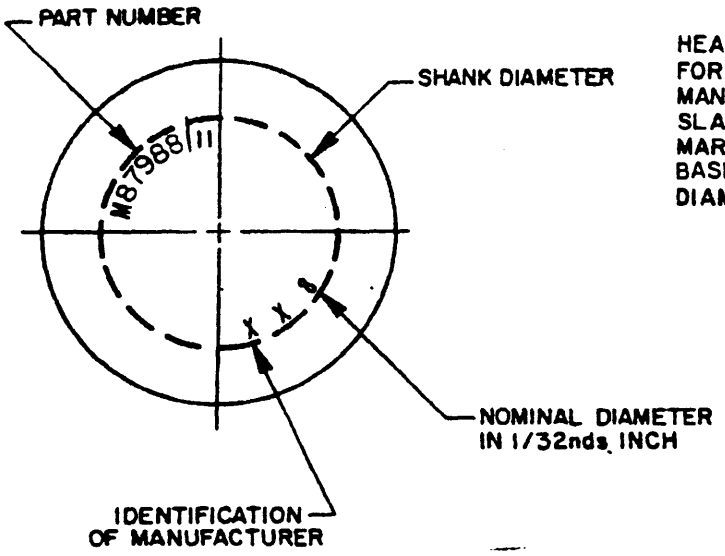
5. Part number. The part number consists of the letter "M", the basic number of this specification sheet, followed by a protective coating and surface treatment code letter, the first number indicates nominal diameters in thirty-seconds of an inch taken from TABLE I, the grip dash number indicates maximum grip length in sixteenths of an inch taken from TABLE II.

Example: M87988/11A8-8



M87988/11A8-8 indicates: Pin-Rivet, Threaded, Type II (100° Flush Reduced Shear Head), Class 2 (6AL-4V Titanium, 95 KSI Fsu and 160 KSI Ftu); aluminum coated; cetyl alcohol lubricant; 8/32 nominal diameter; 8/16 maximum grip length.

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HEAD MARKING SHALL BE .010 MAXIMUM DEEP. FOR DASH 5,6 AND 8 SIZES ONLY, MARK MANUFACTURER IDENTIFICATION AND M PLUS SLASH SHEET (M11). FOR DASH 10 AND ABOVE, MARK MANUFACTURER IDENTIFICATION PLUS BASIC PART NUMBER (M87988/11-) AND DIAMETER.

TABLE II. GRIP AND LENGTH DIMENSIONS

GRIP DASH NO.	MAX GRIP ±.005	LENGTH ±.010[a/]						
		DIAMETER DASH NUMBER AND THREAD SIZE						
		5 .1640-32	6 .1900-32	8 .2500-28	10 .3125-24	12 .3750-24	14 .4375-20	16 .5000-20
1	.062	.375						
2	.125	.437	.450	.520	.625	.670	.760	.810
3	.188	.500	.513	.583	.688	.733	.823	.873
4	.250	.562	.575	.645	.750	.795	.885	.935
5	.312	.625	.638	.708	.813	.858	.948	.998
6	.375	.687	.700	.770	.875	.920	1.010	1.060
7	.438	.750	.763	.883	.938	.984	1.073	1.123
8	.500	.812	.825	.895	1.000	1.045	1.135	1.185
9	.562	.875	.888	.958	1.063	1.108	1.198	1.248
10	.625	.937	.950	1.020	1.125	1.170	1.260	1.310
11	.688	1.000	1.013	1.083	1.188	1.233	1.323	1.373
12	.750	1.062	1.075	1.145	1.250	1.295	1.385	1.435
13	.812	1.125	1.138	1.208	1.313	1.358	1.448	1.498
14	.875	1.187	1.200	1.270	1.375	1.420	1.510	1.560
15	.938	1.250	1.263	1.333	1.438	1.483	1.573	1.623
16	1.000	1.312	1.325	1.395	1.500	1.545	1.635	1.685
17	1.062	1.375	1.388	1.458	1.563	1.608	1.698	1.748
18	1.125	1.437	1.450	1.520	1.625	1.670	1.760	1.810
19	1.188	1.500	1.513	1.583	1.688	1.733	1.823	1.873
20	1.250	1.562	1.575	1.645	1.750	1.795	1.885	1.935
21	1.312	1.625	1.638	1.708	1.813	1.858	1.948	1.998
22	1.375	1.687	1.700	1.770	1.875	1.920	2.010	2.060
23	1.438	1.750	1.763	1.833	1.938	1.983	2.073	2.123
24	1.500	1.812	1.825	1.895	2.000	2.045	2.135	2.185
25	1.562	1.875	1.888	1.958	2.063	2.108	2.198	2.248
26	1.625	1.937	1.950	2.020	2.125	2.170	2.260	2.310
27	1.688	2.000	2.013	2.083	2.188	2.233	2.323	2.373
28	1.750	2.062	2.075	2.145	2.250	2.295	2.385	2.435
29	1.812	2.125	2.138	2.208	2.313	2.358	2.448	2.498
30	1.875	2.187	2.200	2.270	2.375	2.420	2.510	2.560
31	1.938	2.250	2.263	2.333	2.438	2.483	2.573	2.623
32	2.000	2.312	2.325	2.395	2.500	2.545	2.635	2.685

a/ Nominal length equals maximum grip plus "B". See Figure 1.

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Navy - AS
Air Force - 99

Review activities:
Army - GL
DLA - IS
NSA - NS

User activities:

Preparing activity:
Air Force - 82

Agent:
DLA - IS

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